PLASMA PROTEIN BINDING

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
%	Percentage
°C	Degrees Celsius
µL	Microlitre
µM	Micromolar
µm	Micrometre
DMPK	Drug metabolism and pharmacokinetics
DMSO	Dimethyl sulfoxide
DPBS	Dulbecco's phosphate buffer saline
g	Gram
HCl	Hydrochloric acid
HEPES	N-2-hydroxyethylpiperazine-N'-2-ethanesulfonic acid
HOD	Head of the department
hr	Hour
IPM	Intellectual property management
LC-MS/MS	Liquid chromatography coupled with tandem mass spectrometry
LNB	Lab note book
M	Molarity
mAU	Milli absorbance unit
mg	Milligram
min	Minutes
mL	Millilitre
mM	Millimolar
MS	Microsoft
N	Normality
Na_2HPO_4	Disodium hydrogen phosphate
NaH_2PO_4	Sodium di hydrogen phosphate
NaOH	Sodium hydroxide
pH	Potential of hydrogen
PPE	Personal protective equipment
PVDF	Polyvinylidene di fluoride
RED	Rapid equilibrium dialysis
rpm	Revolutions per minute
S.No.	Serial number

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1. OBJECTIVE

The objective of this experiment is to estimate the plasma protein binding potential of test compound(s) following equilibrium dialysis method

2. SCOPE

The procedure described in this protocol will be used for the assessment of plasma protein binding of test compound(s) in rat, mouse, dog, mini pig and human plasma.

3. RESPONSIBILITY

All personnel performing this study are responsible to follow the process outlined in this protocol, at Aurigene Pharmaceutical Services Ltd, India.

4. INTRODUCTION

The pharmacokinetic and pharmacodynamic properties of drugs are largely influenced by the reversible binding of drugs to plasma or serum proteins, such as albumin, glyco-proteins and lipoproteins. In general, only the unbound drug is available for diffusion or transport across cell membranes, for transformation by metabolizing enzymes and for interaction with a pharmacological target. As a result, the extent of plasma protein binding of a drug influences its action as well as its distribution and elimination.

Highly plasma protein bound drugs are confined to the vascular space, thereby having a relatively low volume of distribution. In contrast, drugs that remain largely unbound in plasma are generally available for distribution to other organs and tissues, resulting in large volumes of distribution.

The binding of drugs to proteins both in the vascular space and/ or the extravascular space results in a decrease in drug clearance and a prolonged drug half-life. Only the unbound drug is available for glomerular filtration and in some cases, hepatic clearance.

Equilibrium dialysis is an accurate and reliable method for determining protein binding affinity of chemical or biological substances of low molecular weights.

5. SAFETY AND PRECAUTIONS

During the conduct of the study, the study personnel should take necessary safety precautions and use personal protective equipment (PPE) such as facemasks, gloves, lab coat, goggles, shoes etc.

DMSO:

- Keep away from sources of ignition.
- Do not breathe vapour or mist.
- Avoid contact with skin, eyes and clothing.

6. MATERIALS AND EQUIPMENT

6.1 EQUIPMENT

Equipment	Make
Adhesive permeable seals	Thermo Scientific
Analytical balance	Sartorius/ Denver
Centrifuge with plate/ tube rotors	Eppendorf/ Kubota & Thermo

Equipment	Make
CO_2 Incubator	Thermo Scientific
Cyclomixer	IKA® MTS
Deep freezer (-80°C)	Thermo Scientific/ Sanyo
Deep well plate (96-well)	Axygen scientific
LC-MS/MS (API-5500, API- 4000, API- 3200)	AB Sciex
Micro centrifuge tubes	Tarsons/ Phenomenex
pH meter	Sartorious/ Lab India
Pipettes (10, 100, 200 and 1000 µL)	Eppendorf/ Brand
Plate shaker	IKA® MTS
RED device Inserts	Thermo Scientific
Refrigerator	LG
Software	Microsoft excel
Teflon® base plate	Thermo Scientific
Vortex	IKA®Vortex

Note:

- Equivalent equipment/ instruments could be used based on the availability.

6.2 CHEMICALS

Chemicals	Make; Catalogue No.
Bepridil Hydrochloride	Sigma-Aldrich; B5016
Diethylstilbestrol	J.T.Baker; 9012-03
Dimethyl sulfoxide	Sigma-Aldrich; D8418
Disodium hydrogen phosphate	Sigma-Aldrich; 71640
DPBS	Sigma-Aldrich; D8537
Ethanol	Changshu Hong sheng fine chemicals Cooperation, Limited.
Glyburide	Sigma Aldrich; G2539
Sodium phosphate dibasic	Sigma-Aldrich; S9390
Sodium phosphate monobasic	Sigma-Aldrich; S9638

Note:

- Complete list of chemicals used and their details should be provided in the LNB.
- Equivalent grade chemicals/ reagents from alternate sources can be used on need basis.
- Plasma can be procured from certified vendors for human, rat, mouse, dog & minipig with COA.

- Source information of plasma should be provided along with results

6.3 PREPARATION OF SOLUTIONS AND STORAGE CONDITIONS

S. No.	Composition	Storage condition
1	Preparation of test/ reference compound(s) Prepare 10 mM stock solution in DMSO. Further dilute the stock in DMSO to obtain 0.5 mM solution and subsequently in plasma to obtain a 5 µM working solution maintaining the organic solvent ≤ 1%.	RT Working solution: Prepare fresh
2	Preparation of equilibrium dialysis buffer (0.1 M) Solution A: Add 27.6 g of $NaH_2PO_4.2H_2O$ in 1000 mL of de-ionized water to obtain 0.2 M solution. Solution B: Add 53.6 g of $Na_2HPO_4.7H2O$ in 1000 mL of de-ionized water to get 0.2 M solution. Mix 19 mL of solution A with 81 mL of solution B and adjust the pH to 7.4 using 1 N NaCl/ NaOH. Make up the volume to 1000 mL with de-ionized water.	2-8°C Use within one month
3	Preparation of 20 % v/v ethanol Add 200 mL of ethanol in 800 mL of de-ionized water (v/v) to obtain 20% ethanol.	Prepare fresh
4	Preparation of stop solution Prepare 1 mg/mL stock solution of Glyburide (internal standard) and dilute with acetonitrile to obtain appropriate concentration.	2-8°C

Note:

- Changes to the above mentioned solution method preparation should be captured in the LNB & report (If any).

7. PROCEDURE

Rinse the Teflon® base plate wells with 20% ethanol for 10 min.

Remove alcohol and rinse twice with de-ionized water and allow the plate to dry for few min.

The assay is to be carried out in triplicates (n=3).

Pre-warm the dialysis buffer (100 mM Na_2PO_4, pH 7.4) in an incubator at 37°C.

Thaw the plasma at 37°C and centrifuge the plasma for 15 min at 3500 rpm at 4°C.

Prepare the test and reference compounds as above.

For recovery, draw 50 µL plasma samples from the working solution prior to incubation (T_0 hr) and dispense in to separate wells in a deep well plate (96-well). Add 50 µL of buffer and mix well to normalize the matrix.

Place inserts (open end up) into the wells of the Teflon® base plate.

Add 300 µL of working solution (plasma sample) to the plasma chamber indicated by the red ring and 500 µL of dialysis buffer to the buffer chamber.

Seal the unit with plate sealer and incubate at 37°C on an orbital shaker at 450 rpm, 5% CO_2 for 4 hr.

Remove plate sealer and take out 50 µL from both buffer and plasma chambers and dispense into separate wells in a 96 well-deep well plate.

Add 50 µL of blank plasma to the buffer samples and 50 µL of buffer to the plasma samples and mix well to normalize the matrix.

To the above wells add 300 µL of stop solution and centrifuge the samples for 15 min at 3500 rpm at 4°C.

Take out the supernatant, transfer into LC-MS/MS vials and submit for analysis.

Discard RED inserts, wash the Teflon® base plate with 20% ethanol, dry and keep for reuse.

Note:

- Modifications to the procedure, if any, should be captured in the LNB & report.
- Appropriate bio-analytical method will be used for the quantification of the analyte

8. ASSAY CONTROL

Bepridil should be included as assay control.

Study would be qualified based on the acceptance criteria given in the data template.

Note:

- Any changes to the assay controls would be made only with the consent of the sponsor and should be described in the LNB & report ((If any).

9. DATA RECORDING

The data should be tabulated in MS excel based plasma protein binding assay template and the final data should be pasted in the LNB. Raw data should be recorded in the LC-MS/MS server with the study number and the data path should be provided in the excel file and LNB.

10. DATA ANALYSIS

Peak area ratio (test compound peak area/ internal standard peak area) in the plasma and buffer samples can be used to determine percent bound. Determine recovery of the test compound in plasma using (T0 hr) sample. Calculate the percentage bound, percentage unbound and percentage recovery by using the formula given below:

$$\% \text{ free} = \frac{\text{Peak area ratio in buffer sample}}{\text{Peak area ratio in plasma sample}} \times 100$$

$$\text{fu (Fraction bound)} = 100 - \% \text{ free}$$

$$\% \text{ Recovery} = \frac{\text{Peak area ratio in buffer (4 hr)} + \text{Peak area ratio in plasma (4 hr)}}{\text{Peak area ratio in plasma (0 hr)}} \times 100$$

$$\text{LOQ } \%\text{fu} = \frac{3\text{x Blank}}{\text{Average } (T_0 \text{ area plasma})} \times 100$$

11. REPORTING

Data should be reported in an MS excel format. Based on the need, MS word based reports may also be prepared.

If reference compound(s) does not meet the specified acceptance criteria, the complete batch should be rejected.

Condition	Report Status	Remark
Precipitation observed in tube	No data reported	Precipitation in DMSO stock solution
No peak present in plasma at time=0	Undefined	Test compound not detectable in samples
Plasma time=0 peak area < lower limit	No data reported	Analytical issue - no MS/too low signal
Poor peak shape observed and accuracy of result compromised	No data reported	Analytical issue - Poor chromatogram
% free <1 %	Report as <1%	N/A
% free >80 % (very highly unbound compounds)	Report % free as >80%	N/A
Recovery between <55 % or >130 %	No data reported	Recovery outside of acceptance criteria
CV(%free n=3) >50%	if CV(% free n=2) <50% , report % free calculated with 2 replicates	% free calculated with only 2 replicates
CV(%free n=3) >50%	if CV(% free n=2) >50%, do not report any results for %free	Undefined - High variability in % free
% free >1% AND % free < LOQ% free	No data reported	LOQ issue

12. DATA ARCHIVAL

All the raw data, chromatograms and reports should be stored in the server at Aurigene Pharmaceutical Services Limited, India for a period of five years or as specified by sponsor. Assay details and analytical procedures should be captured in LNB and the same should be archived as per the instructions/ procedure of Intellectual Property Management (IPM) department.

13. REFERENCES

- http://www.cambridgemedchemconsulting.com/resources/ADME/distribution.html.
- Wright JD, Boudinot FD, Ujhelyi MR; Measurement and analysis of unbound drug concentrations; Clinical Pharmacokinetics; Jun; 30 (6); 445-62; 1996.

14. CHANGE AND AMENDMENT HISTORY

Change History		
Version	**Changes made/ Reason for changes**	**Effective date**
2.0	Revised Version	1 Jul 2022